

TinyBolus Formulary

Peer-review document

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Source: github.com/SGP-Prime/tinybolusformulas

Privacy policy: sgp-prime.github.io/tinybolus-site/privacy.html

This is a reference tool, not a prescribing system. Verify every value against your hospital protocols, product labelling, and the individual patient in front of you before administration.

About this document

This is the human-readable export of the TinyBolos paediatric reference formulary, organised section-by-section as it appears in the application. Each item is presented with its dose specification, applicable age brackets (where present), route, clinical notes, and numerical references pointing into the bibliography at the end of the document.

How to read each item

Items fall into a small number of shapes that map directly onto how the application computes them:

- **Single dose** (e.g., 10 µg/kg): one factor multiplied by weight, often with a hard maximum cap.
- **Dose range** (e.g., 2–3 mg/kg): a low–high range, also weight-scaled, optionally with a maximum cap on either bound.
- **Age-bracketed**: different dose factors for different age cohorts. Rendered as a small table per item.
- **Age-bracketed fixed value**: not weight-scaled, just a string per age band (e.g., laryngoscope blade size).
- **Computed**: piecewise logic that doesn't reduce to a single formula (ETT sizing, fluid maintenance, LMA / supraglottic airway sizing, vital-sign normal ranges).

Hard floors / contraindications

Some items carry an *age* or *weight* floor below which the formulary explicitly declines to suggest a dose. In the app this renders as ∅ (empty-set glyph) with a "not indicated" explanation. Items with such floors are flagged in red below each row.

References

Every dose / range / age bracket / sizing rule carries one or more citations. References are numbered in encounter order throughout this document, with a full bibliography at the end. The numbers in [brackets] after each item refer to entries in that bibliography.

Source

The formulary itself lives at github.com/SGP-Prime/tinybolusformulas as a single machine-readable JSON file. This document is regenerated from that file on demand by `tool/generate_formulary_review_html.dart`; do not hand-edit it.

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EMERGENCY — PALS

Compression rate

VALUE 100–120/min

References: [\[1\]](#)

Compression depth

RULE 4 cm <1 y · 5 cm ≥1 y

not exceeding 6 cm at any age

References: [\[1\]](#)

C:V ratio

VALUE 15:2

15:2 if PBLs-trained · 30:2 if untrained or adult-BLS-only (ERC 2025 – split is by training, not rescuer count)

References: [\[1\]](#)

Defibrillation

DOSE 4 J/kg

increase stepwise up to 8 J/kg (max 360 J) for refractory VF/pulseless VT from the 6th shock · adolescents: adult first-shock energy 120–200 J

References: [\[1\]](#), [\[2\]](#)

Cardioversion (synchronised)

DOSE 1 J/kg

double per attempt to max 4 J/kg

References: [\[1\]](#)

Adrenaline (cardiac arrest)

IV/IO

DOSE 10 µg/kg (max 1 mg)

repeat every 3–5 min · max 1 mg/dose

References: [\[1\]](#), [\[3\]](#)

Amiodarone

IV/IO

DOSE 5 mg/kg (max 300 mg)

after the 3rd shock · repeat after the 5th (max 150 mg subsequent doses) (ERC 2025)

References: [\[1\]](#)

Atropine

IV

DOSE 20 µg/kg (max 500 µg)

20 µg/kg · max single dose 500 µg (ERC 2025) · no minimum dose (Barrington 2011)

References: [\[1\]](#), [\[4\]](#), [\[5\]](#)

Adrenaline (bradycardia)

IV/IO

DOSE 1–2 µg/kg

bradycardia unresponsive to oxygenation/ventilation · atropine if vagal cause (ERC 2025)

References: [\[1\]](#), [\[4\]](#)

Adenosine (1st dose)

IV

DOSE 0.1 mg/kg (max 6 mg)

References: [\[1\]](#)

Adenosine (2nd dose)

IV

DOSE 0.3 mg/kg (max 18 mg)

if SVT persists, escalate in 0.05–0.1 mg/kg steps to a max single dose of 0.5 mg/kg (ERC 2025)

References: [\[1\]](#)

Magnesium sulphate (torsades)

IV/IO

DOSE 50 mg/kg (max 2 g)

torsade-de-pointes VT (ERC 2025)

References: [\[1\]](#)

Vital Signs

Heart Rate

≤1 mo	120–170 bpm
2–12 mo	100–160 bpm
1–2 y	98–140 bpm
3–5 y	80–120 bpm
6–11 y	75–118 bpm
≥12 y	60–100 bpm

References: [\[6\]](#)

Systolic BP

≤12 mo	72–104 mmHg
1–2 y	86–106 mmHg
3–5 y	89–112 mmHg
6–9 y	97–115 mmHg
10–11 y	102–120 mmHg
12–15 y	110–131 mmHg
≥16 y	90–140 mmHg

References: [\[6\]](#)

Diastolic BP

≤12 mo	37–56 mmHg
1–2 y	42–63 mmHg
3–5 y	46–72 mmHg
6–9 y	57–76 mmHg
10–11 y	61–80 mmHg
12–15 y	64–83 mmHg
≥16 y	60–90 mmHg

References: [\[6\]](#)

Respiratory Rate

≤1 mo	40–60 /min
2–12 mo	30–53 /min
1–3 y	22–37 /min
4–5 y	20–28 /min
6–12 y	18–25 /min
≥13 y	12–20 /min

References: [\[6\]](#)

Tidal Volume

DOSE 5–7 mL/kg

per IDEAL body weight – in obesity use age-estimated, not actual, weight · guidelines target 6–8 mL/kg IBW · lower end for neonates

References: [\[7\]](#), [\[8\]](#)

Induction

Propofol

IV

≤1 mo	1–2.5 mg/kg
2–12 mo	3–4 mg/kg
1–6 y	3–3.5 mg/kg
7–12 y	2.5–3 mg/kg
≥13 y	2–2.5 mg/kg

after a sevoflurane inhalational induction, roughly halve the dose (lower requirement)

References: [\[9\]](#), [\[10\]](#), [\[11\]](#), [\[12\]](#)

Ketamine

IV

DOSE 1–2 mg/kg

References: [\[13\]](#), [\[14\]](#)

Ketamine (IM)

IM

DOSE 4–5 mg/kg

IM route when IV access impractical · onset 3–5 min · full anaesthesia may require up to 10 mg/kg (SmPC)

References: [\[13\]](#), [\[14\]](#)

Rocuronium

IV

DOSE 0.6–1.2 mg/kg

intubation 0.6 · RSI 1.0–1.2

References: [\[15\]](#), [\[16\]](#)

Succinylcholine (RSI)

IV

≤12 mo 2–3 mg/kg — *raised in infants — larger extracellular-fluid volume of distribution (Meakin 1989)*

≥1 y 1–2 mg/kg

intubating dose · laryngospasm + IM doses in Respiratory · contraindicated: MH susceptibility, hyperkalaemia risk

References: [\[16\]](#), [\[6\]](#), [\[17\]](#)

Fentanyl (induction)

IV

≤1 mo 1–3 µg/kg — *neonate: reduce dose · inject slowly (chest-wall rigidity) · monitor for apnoea (SPA 2019)*

≥2 mo 1–3 µg/kg

References: [\[14\]](#), [\[18\]](#)

Midazolam

IV

≤6 mo 0.1 mg/kg — *young infant <6 mo: reduced clearance — prolonged effect · slow injection · hypotension, esp. with an opioid*

≥7 mo 0.1 mg/kg

CAP max 5 mg

References: [\[14\]](#)

Midazolam (oral premed)

PO

DOSE 0.5 mg/kg (max 20 mg)

References: [\[14\]](#)

Midazolam (buccal premed)

BUCCAL

DOSE 0.3 mg/kg (max 10 mg)

buccal alternative to oral premed (e.g. swallowing refused) · onset ~10–15 min · (for seizures see Emergency)

References: [\[14\]](#)

Midazolam (IN premed)

IN

DOSE 0.2–0.3 mg/kg (max 10 mg)

intranasal premedication · 0.3 mg/kg for faster, more reliable onset · 5 mg/mL solution – divide between nostrils

References: [\[19\]](#), [\[14\]](#)

Ondansetron

IV

DOSE 0.1 mg/kg (max 4 mg)

Contraindicated <1 months

References: [\[20\]](#), [\[14\]](#), [\[21\]](#)

Dexamethasone (PONV)

IV

DOSE 0.15–0.25 mg/kg (max 4 mg)

PONV prophylaxis at induction · 0.25 mg/kg effective ceiling (no benefit above)

References: [\[20\]](#), [\[22\]](#), [\[14\]](#)

Dexamethasone (anti-inflammatory)

IV

DOSE 0.5 mg/kg (max 8 mg)

airway oedema / post-extubation stridor · 0.5 mg/kg/dose (vs the 0.15 mg/kg PONV dose above)

References: [\[23\]](#), [\[24\]](#)

Etomidate

IV

DOSE 0.2–0.3 mg/kg

haemodynamically stable induction

References: [\[14\]](#), [\[6\]](#)

Atracurium

IV

DOSE 0.4–0.5 mg/kg

intubating dose · 0.3–0.4 mg/kg in infants

References: [\[14\]](#), [\[6\]](#)

Dexmedetomidine (IN premed)

IN

DOSE 1–3 µg/kg (max 200 µg)

intranasal premedication · 2 µg/kg usual · onset ~30 min · 100 µg/mL solution

References: [\[25\]](#), [\[26\]](#)

Thiopental

IV

≤1 mo 3–4 mg/kg

2–12 mo 5–8 mg/kg

≥1 y 5–6 mg/kg

then 1 mg/kg PRN · titrate to effect – SmPC: 'no fixed dosage' · adults 4–6 mg/kg

References: [\[27\]](#)

Painkillers

Paracetamol

IV

≤10 kg 7.5 mg/kg

>10 kg 15 mg/kg

CAP max 1 g

IV dosing splits by weight, not age (SmPC): ≤10 kg → 7.5 mg/kg · interval ≥4 h · daily max 30 mg/kg
≤10 kg / 60 mg/kg >10 kg (SmPC)

References: [\[28\]](#), [\[14\]](#), [\[29\]](#)

Ketorolac

IV

DOSE 0.5 mg/kg (max 30 mg)

Contraindicated <6 months

References: [\[30\]](#), [\[14\]](#)

Morphine

IV

≤1 mo 0.03–0.05 mg/kg

≥2 mo 0.1 mg/kg

References: [\[31\]](#), [\[14\]](#)

Fentanyl (analgesia)

IV

≤1 mo 0.5–1 µg/kg — *neonate: reduce dose · inject slowly · monitor for apnoea (SPA 2019)*

≥2 mo 0.5–1 µg/kg

References: [\[14\]](#), [\[32\]](#), [\[18\]](#)

Metamizole

IV

≤11 mo 5 mg/kg — *reduced in infants <1 y — higher exposure per dose from immature metabolism (Ziesenitz 2019)*

≥1 y 10–20 mg/kg

CAP max 1 g

Contraindicated <3 months

References: [\[33\]](#), [\[34\]](#)

Ibuprofen (IV)

IV

DOSE 10 mg/kg (max 400 mg)

Contraindicated <3 months

not <3 mo or <5 kg · infuse over ≥10 min · IV ibuprofen for preterm PDA closure is a separate indication/course

References: [\[35\]](#), [\[14\]](#), [\[36\]](#), [\[37\]](#), [\[38\]](#)

Ibuprofen (oral)

PO

DOSE 5–10 mg/kg (max 400 mg)

Contraindicated <3 months

analgesic / antipyretic · typically 7–10 mg/kg/dose · not <3 mo or <5 kg

References: [\[35\]](#), [\[14\]](#), [\[36\]](#)

Tramadol

IV/PO

DOSE 1–2 mg/kg (max 100 mg)

Contraindicated <12 months

q6h (max 4×/day) · daily max 8 mg/kg or 400 mg, whichever lower (SmPC) · respiratory-depression caution · oral-drops licence: confirm Infarmed RCM

References: [\[39\]](#), [\[40\]](#)

Diclofenac (PO/PR)

PO/PR

DOSE 0.3–1 mg/kg

Contraindicated <12 months

q8–12h · daily total 1–3 mg/kg (max 200 mg/day) · post-op course ≤4 d (SmPC) · NL formulary licenses from 3 mo

References: [\[41\]](#), [\[40\]](#)

Paracetamol (oral)

PO

DOSE 15 mg/kg (max 1 g)

q4–6h (max 4×/day) · daily max 60 mg/kg (max 4 g) · neonates: max 3×/day, 45 mg/kg/day · short-term post-op up to 90 mg/kg/day ≤3 d (Kinderformularium)

References: [\[40\]](#)

Emergency Drugs

Adrenaline (cardiac arrest)

IV/IO

DOSE 10 µg/kg (max 1 mg)

repeat every 3–5 min · max 1 mg/dose

References: [\[1\]](#), [\[3\]](#)

Adrenaline (anaphylaxis IM)

IM

DOSE 10 µg/kg (max 500 µg)

weight-based 10 µg/kg (shown) · fixed bands: <6 mo 100–150 µg (RCUK) · 1–5 y 150 µg · 6–12 y 300 µg · >12 y 500 µg (ERC 2025) · repeat every 5 min if needed

References: [\[1\]](#), [\[4\]](#)

Adrenaline (anaphylaxis IV)

IV

DOSE 1 µg/kg

titrate to response · IM adrenaline is first-line

References: [\[1\]](#)

Atropine

IV

DOSE 20 µg/kg (max 500 µg)

20 µg/kg · max single dose 500 µg (ERC 2025) · no minimum dose (Barrington 2011)

References: [\[1\]](#), [\[4\]](#), [\[5\]](#)

Ephedrine

IV

≤6 mo 0.6–1.2 mg/kg — *young infants need ~10× the older-child dose – indirect agent, immature sympathetic system · 1.2 mg/kg optimal (Szostek 2023)*

≥7 mo 0.1–0.3 mg/kg

ampoule 30 mg/mL – dilute before paediatric use

References: [\[42\]](#), [\[14\]](#), [\[6\]](#)

Phenylephrine

IV

DOSE 5–20 µg/kg (max 500 µg)

5–20 µg/kg spans indications – upper range for tet spells · for hypotension start low and titrate

References: [\[14\]](#), [\[6\]](#)

Lidocaine

IV

DOSE 1 mg/kg (max 100 mg)

antiarrhythmic – alternative to amiodarone for shock-refractory VF/pVT

References: [\[1\]](#), [\[3\]](#)

Glucose 10%

IV/IO

DOSE 2 mL/kg

0.2 g/kg for hypoglycaemia · recheck glucose after

References: [\[1\]](#), [\[43\]](#)

Calcium gluconate 10%

IV/IO

DOSE 0.5 mL/kg (max 20 mL)

hypocalcaemia · hyperkalaemia with ECG changes – but not in cardiac arrest (ERC 2025) · calcium-channel-blocker toxicity · give slowly over 5–10 min

References: [\[1\]](#), [\[43\]](#)

Sodium bicarbonate 8.4%

IV/IO

DOSE 1 mmol/kg

1 mL/kg of 8.4% · tricyclic (TCA) toxicity or severe acidosis per local protocol · NOT for hyperkalaemic arrest (ERC 2025: avoid calcium & bicarbonate) · not routine in arrest

References: [\[1\]](#), [\[43\]](#)

Dantrolene

IV

DOSE 2.5 mg/kg

initial 2–2.5 mg/kg · then 1 mg/kg every 5 min as needed · 10 mg/kg is a reappraisal point – may be exceeded (EMHG) · reconstitute each 20 mg vial in 60 mL water – start reconstituting immediately

References: [\[44\]](#), [\[45\]](#)

Midazolam (buccal, status epilepticus)

BUCCAL

 ≤ 11 mo 2.5 mg

12 mo – 4 y 5 mg

5–9 y 7.5 mg

 ≥ 10 y 10 mg**Contraindicated <3 months**

convulsive seizure ≥ 5 min · 3–6 mo: hospital setting · repeat once after 5–10 min if persisting, then levetiracetam 40–60 mg/kg (max 4.5 g) IV/IO (ERC 2025)

References: [\[46\]](#), [\[1\]](#)

Midazolam (IV, status epilepticus)

IV/IO

DOSE 0.15 mg/kg (max 10 mg)

convulsive SE with IV/IO access already in place · otherwise buccal route first · may repeat once after 5–10 min · off-label: injection SmPCs cover sedation/anaesthesia only

References: [\[47\]](#), [\[48\]](#), [\[1\]](#)

Hydrocortisone (adrenal crisis)

IV

 ≤ 1 mo 4 mg/kg ≥ 2 mo 2 mg/kg

CAP max 100 mg

adrenal crisis – then 1 mg/kg (max 50 mg) 6-hourly, neonate 2 mg/kg (BSPED) · give 0.9% saline 10 mL/kg bolus · glucose 2 mL/kg 10% if hypoglycaemic · age-band alt: 25 mg <1 y / 50 mg 1–5 y / 100 mg ≥ 6 y · anaphylaxis: only when asthma-associated (ERC 2025)

References: [\[49\]](#), [\[1\]](#)

Insulin + glucose (hyperkalaemia)

IV

DOSE 0.1 units/kg (max 10 units)

soluble insulin 0.1 units/kg WITH glucose 10% 5 mL/kg (0.5 g/kg, max 250 mL) – never insulin without glucose · check glucose & K⁺ every 15 min for 4 h · follow with glucose infusion (ERC 2025)

References: [\[50\]](#), [\[1\]](#)

Salbutamol (nebulized, hyperkalaemia)

NEBULIZED

VALUE 2.5–5 mg

2.5 mg ≤ 25 kg · 5 mg > 25 kg (RCH; asthma row is age-banded) · repeat up to 5× (ERC 2025) · adjunct alongside insulin–glucose

References: [\[50\]](#), [\[1\]](#)

Fluids and Blood

Fluid Bolus

IV

DOSE 10 mL/kg

balanced isotonic crystalloid · cardiogenic shock: 5 mL/kg · adolescents: max 500 mL per bolus (ERC 2025) · reassess after each

References: [\[1\]](#)

Maintenance

RULE Holliday-Segar 4-2-1: 4 mL/kg/h (first 10 kg) · 2 mL/kg/h (next 10 kg) · 1 mL/kg/h thereafter

4-2-1 rule (Holliday-Segar) · isotonic fluid + glucose ± K⁺ (Feld 2018) · <1 mo: neonatal mL/kg/day regimen applies, not 4-2-1

References: [\[51\]](#), [\[52\]](#)

Total Blood Volume

≤1 mo 85 mL/kg

2 mo – 2 y 75 mL/kg

≥3 y 70 mL/kg

References: [\[6\]](#)

Tranexamic Acid

IV

DOSE 15 mg/kg (max 1 g)

loading dose over 10 min · then 2 mg/kg/h infusion (RCPCH)

References: [\[53\]](#), [\[54\]](#), [\[55\]](#)

Max allowable blood loss

VALUE $EBV \times (Hct - \text{min Hct}) / Hct$

estimate only – never a standalone transfusion trigger (SPA) · conservative form (initial-Hct denominator); Gross's correction divides by MEAN Hct · EBV: preterm 90–100 · neonate 80–90 · infant 70–80 · >1 y 70 mL/kg

References: [\[56\]](#), [\[57\]](#)

Equipment

ETT

RULE Cuffed: $3.5 + \text{age}/4$, rounded to ± 0.5

<1 MO Neonatal weight table (≤ 4 kg, uncuffed):

≤ 1.5 kg size 2.5

≤ 3 kg size 3.0

≤ 4 kg size 3.5

INFANT age table (cuffed):

≤ 6 mo size 3.5

7–12 mo size 4.0

formula & infant table are cuffed sizes – uncuffed: add 0.5 (≥ 1 y) · neonatal table (<1 mo) shows uncuffed sizes

References: [\[58\]](#), [\[59\]](#)

ETT insertion depth

RULE $6 + \text{weight (<1 mo)} \cdot \text{weight}/2 + 8$ (<1 y) · $\text{age}/2 + 12$ (≥ 1 y) (cm)

nasal: add 1 cm <1 y · 2–3 cm ≥ 1 y · ≥ 1 y oral formula per APLS (Lau's data suggest $\text{age}/2 + 13$)

References: [\[60\]](#), [\[61\]](#), [\[43\]](#)

iGel LMA

IGEL SIZING weight bands (inclusive on both bounds):

2–5 kg size 1

5–12 kg size 1.5

10–25 kg size 2

25–35 kg size 2.5

30–60 kg size 3

50–90 kg size 4

90– ∞ kg size 5

References: [\[62\]](#), [\[63\]](#)

AuraGain LMA

AURAGAIN SIZING weight bands (inclusive on both bounds):

2–5 kg	size 1
5–10 kg	size 1.5
10–20 kg	size 2
20–30 kg	size 2.5
30–50 kg	size 3
50–70 kg	size 4
70–100 kg	size 5
100–∞ kg	size 6

References: [\[64\]](#)

Laryngoscope blade

≤1 y	Miller 0–1
2–7 y	Macintosh 2
≥8 y	Macintosh 3

infants: straight (Miller) blade lifts the epiglottis directly · Miller and Macintosh size 1 give equivalent views (Passi 2014)

References: [\[65\]](#), [\[6\]](#)

Face mask

≤1 mo	Size 0
2–12 mo	Size 1
1–3 y	Size 2
4–8 y	Size 3
9–14 y	Size 4
≥15 y	Size 5

mask should cover nose and mouth from eyebrow line to chin cleft · anatomical fit is more important than estimated size

References: [\[6\]](#), [\[66\]](#)

Reservoir bag

≤6 mo 0.5 L

7 mo – 6 y 1 L

≥7 y 2 L

anaesthesia breathing-circuit bag · 0.5 L neonate/infant, 1 L to ~6 y (<25-30 kg), 2 L older child and adult

References: [\[67\]](#), [\[68\]](#)

Suction catheter

≤1 mo 6 Fr

2 mo – 1 y 8 Fr

2–8 y 10 Fr

≥9 y 12–14 Fr

References: [\[43\]](#), [\[6\]](#)

NG/OG tube

≤1 mo 6 Fr

2 mo – 1 y 8 Fr

2–8 y 10 Fr

≥9 y 12–14 Fr

gastric decompression may need larger sizes in adolescents

References: [\[43\]](#), [\[6\]](#)

OPA / NPA sizing

VALUE by landmark

OPA (Guedel): central incisors → angle of mandible · NPA depth: nostril → tragus · landmarks only ~40–50% accurate – always confirm clinically (Nemeth) · NPA: avoid in basal skull fracture / coagulopathy (ERC 2021)

References: [\[48\]](#), [\[69\]](#), [\[70\]](#)

Respiratory

Salbutamol (inhaled)

INHALED

≤5 y 2–10 puffs

≥6 y 4–10 puffs

100 µg/puff · repeat incrementally as needed

References: [\[71\]](#), [\[72\]](#)

Salbutamol (nebulized)

NEBULIZED

≤5 y 2.5 mg

≥6 y 5 mg

repeat back-to-back or continuously per response

References: [\[71\]](#), [\[72\]](#)

Ipratropium (nebulized)

NEBULIZED

≤11 y 250 µg

≥12 y 500 µg

References: [\[14\]](#)

Magnesium Sulphate

IV

DOSE 25–50 mg/kg (max 2 g)

severe asthma · infuse over 20 min (ERC 2025)

References: [\[73\]](#), [\[14\]](#), [\[1\]](#)

Clemastine

IV

DOSE 25 µg/kg (max 2 mg)

Contraindicated <12 months

antihistamine adjunct (anaphylaxis / allergy) – second-line, after adrenaline · SmPC: not <1 y

References: [\[74\]](#), [\[75\]](#), [\[76\]](#)

Adrenaline (nebulized)

NEBULIZED

DOSE 0.5 mg/kg (max 5 mg)

croup / post-extubation stridor · 1:1000 solution (1 mg/mL) · repeat per response

References: [\[77\]](#), [\[14\]](#)

Propofol (laryngospasm)

IV

DOSE 0.5 mg/kg

drug of choice before succinylcholine – 0.5 mg/kg increments, deepen anaesthesia (Gavel 2014)

References: [\[78\]](#)

Succinylcholine (laryngospasm)

IV

DOSE 0.1–0.5 mg/kg

start low – 0.1 mg/kg may break spasm while preserving ventilation (Gavel 2014) · contraindicated: MH susceptibility, hyperkalaemia risk

References: [\[79\]](#), [\[78\]](#)

Succinylcholine (IM)

IM

DOSE 4 mg/kg (max 200 mg)

for laryngospasm or RSI without IV access · 4 mg/kg (max 200 mg) · contraindicated: MH susceptibility, hyperkalaemia risk

References: [\[79\]](#), [\[78\]](#)

Rocuronium (laryngospasm)

IV

DOSE 0.6–1.2 mg/kg

when succinylcholine is contraindicated (e.g. MH) – full intubating dose, then secure the airway · sugammadex 16 mg/kg available for reversal

References: [\[16\]](#), [\[15\]](#)

Salbutamol (IV, severe asthma)

IV

DOSE 15 µg/kg (max 250 µg)

single bolus over 10 min if refractory to inhaled therapy (SIGN 158/RCUK 2025) · then infusion 1–2 µg/kg/min (HDU); >2, up to 5, PICU only · monitor ECG + K+ · off-label <12 y per SmPC

References: [\[71\]](#), [\[80\]](#)

Prednisolone (PO, asthma)

PO

DOSE 1–2 mg/kg (max 40 mg)

within the first hour · oral = IV efficacy · cap 40 mg (RCUK 2025; ERC 2021: max 60 mg/day; SIGN bands: 10 mg <2 y / 20 mg 2–5 y / 30–40 mg >5 y) · course ≤3 d, no taper (SIGN 158)

References: [\[80\]](#), [\[71\]](#), [\[48\]](#)

Hydrocortisone (IV, asthma)

IV

DOSE 4 mg/kg

repeated 4-hourly · reserve for children unable to retain oral steroids (SIGN 158)

References: [\[71\]](#)

Reversal Agents

Naloxone (titrated)

IV

DOSE 10 µg/kg

post-anaesthesia reversal — titrate to respiration · repeat every 2–3 min

References: [\[14\]](#), [\[1\]](#)

Naloxone (overdose)

IV/IO/IM

DOSE 0.1 mg/kg (max 2 mg)

suspected opioid overdose with apnoea or unresponsiveness

References: [\[1\]](#), [\[14\]](#)

Flumazenil

IV

DOSE 10 µg/kg (max 200 µg)

max bolus 200 µg · repeat up to 4× · total max 50 µg/kg (up to 1 mg) · licensed ≥1 y

References: [\[14\]](#)

Sugammadex

IV

DOSE 2–4 mg/kg

2 mg/kg moderate block · 4 mg/kg deep block

References: [\[81\]](#), [\[82\]](#)

Sugammadex (profound block)

IV

DOSE 16 mg/kg

for can't-intubate-can't-oxygenate (CICO) – immediate reversal of full intubating-dose rocuronium · off-label in children (adult evidence; SmPC: not investigated in paediatrics)

References: [\[83\]](#), [\[84\]](#)

Neostigmine

IV

DOSE 0.04–0.07 mg/kg (max 5 mg)

coadminister with atropine to block muscarinic effects · give slowly over 1–2 min after first signs of spontaneous recovery

References: [\[14\]](#), [\[6\]](#)

Atropine (with neostigmine)

IV

DOSE 20 µg/kg (max 500 µg)

give together with or just before neostigmine to blunt bradycardia and bronchorrhoea

References: [\[14\]](#), [\[6\]](#)

Neostigmine + glycopyrronium

IV

DOSE 0.02 mL/kg (max 2 mL)

combined 2.5/0.5 mg per mL solution: 0.02 mL/kg = neostigmine 50 µg/kg + glycopyrronium 10 µg/kg · give over 10–30 s · do not exceed 2 mL total (SmPC)

References: [\[85\]](#)

Regional Anesthesia

Lidocaine (perineural)

PERINEURAL

≤5 mo maximum 3.5 mg/kg

≥6 mo maximum 5 mg/kg

plain solution · SmPC 2–11 y: 3–4 mg/kg – 5 mg/kg per US practice (Coté)

References: [\[6\]](#), [\[14\]](#)

Bupivacaine (perineural)

PERINEURAL

≤5 mo maximum 1.5–2 mg/kg — *infusion: 0.2 mg/kg/h*

≥6 mo maximum 2–2.5 mg/kg — *infusion: 0.4 mg/kg/h*

References: [\[86\]](#), [\[87\]](#)

Ropivacaine (perineural)

PERINEURAL

≤5 mo maximum 2 mg/kg — *infusion: 0.2 mg/kg/h*

≥6 mo maximum 3 mg/kg — *infusion: 0.4 mg/kg/h*

References: [\[88\]](#), [\[89\]](#), [\[87\]](#)

Intralipid 20% bolus

IV

DOSE 1.5 mL/kg

local-anaesthetic systemic toxicity (LAST) rescue · max total 12 mL/kg

References: [\[86\]](#), [\[90\]](#)

Intralipid 20% infusion

IV

DOSE 0.25 mL/kg/min

LAST rescue – continue until stable

References: [\[86\]](#), [\[90\]](#)

Levobupivacaine (max)

PERINEURAL

DOSE maximum 1.3 mg/kg (max 150 mg)

Contraindicated <6 months

per side – ilioinguinal/iliohypogastric block (SmPC) · no SmPC ceiling established for other paediatric blocks · adult max single dose 150 mg / 24 h 400 mg

References: [\[91\]](#)

Weight Estimation

Used when the user does not supply a measured weight. <1 month and >12 years always require user input.

1–12 mo (infant)

FORMULA $\text{weight (kg)} = (0.5 \times \text{months}) + 4$

References: [\[43\]](#)

1–5 y (young child)

FORMULA $\text{weight (kg)} = (2 \times \text{age}) + 8$

References: [\[43\]](#)

6–12 y (school-age)

FORMULA $\text{weight (kg)} = (3 \times \text{age}) + 7$

References: [\[92\]](#), [\[43\]](#)

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